PROJECT PROPOSAL

Blockchain-Based Enrollment Management System for Colegio de Marinduque

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A. Problem Formulation (PICO(T) Framework)

The current enrollment process at Colegio de Marinduque is characterized by manual, fragmented, and paper-based procedures that result in long queues, redundant verification, delayed processing, and limited traceability of student records. These inefficiencies are further compounded by the absence of a centralized digital platform and weak security mechanisms, exposing records to risks such as tampering, data loss, and unauthorized modification.

Using the **PICO(T) Framework**, the problem is structured as follows:

- **Population (P):** Students, registrar staff, and administrative personnel at Colegio de Marinduque
- **Intervention (I):** Blockchain-based enrollment management system integrated with a web-based platform
- **Comparison (C):** Traditional manual and centralized enrollment systems
- **Outcome (O):** Improved efficiency, data integrity, transparency, and secure record management
- **Time (T):** During enrollment periods per academic semester

Problem Statement (PICO Format)

Can a blockchain-based enrollment management system improve efficiency, data integrity, and transparency in the enrollment process of Colegio de Marinduque compared to traditional manual and centralized systems during each academic enrollment period?

B. Research Questions (FINER Criteria)

The study is guided by the following research questions, aligned with the FINER criteria:

1. How can a blockchain-integrated enrollment system be designed and developed to centralize student registration, subject enrollment, payment verification, and academic record management?
2. How does blockchain technology enhance data security, transparency, and immutability compared to traditional enrollment systems?
3. How effective and acceptable is the proposed system in terms of usability, efficiency, accuracy, security, and overall performance based on ISO/IEC 25010 standards?

These questions are feasible within available tools (MERN stack + blockchain simulation), novel in local institutional context, ethical under RA 10173 compliance, and highly relevant to digital transformation in education.

C. SMART Goals

General Goal

To develop and evaluate a **Blockchain-Based Enrollment Management System** that improves efficiency, security, and transparency of enrollment processes at Colegio de Marinduque.

Specific SMART Goals

1. **Specific:**

   Develop a web-based enrollment system with blockchain hashing for secure record management.
2. **Measurable:**
   - Achieve API response time ≤ **350ms**
   - Maintain system accuracy ≥ **98%**
   - Support at least **250 concurrent users**
3. **Achievable:**

   Utilize MERN stack, Hyperledger Fabric blockchain simulation, and cloud deployment tools (Vercel, MongoDB Atlas).
4. **Relevant:**

   Addresses real institutional problems: inefficiency, data insecurity, and lack of traceability.
5. **Time-bound:**

   Complete development, testing, and evaluation of the minimum viable product (MVP) within 1 month.

Deliverables

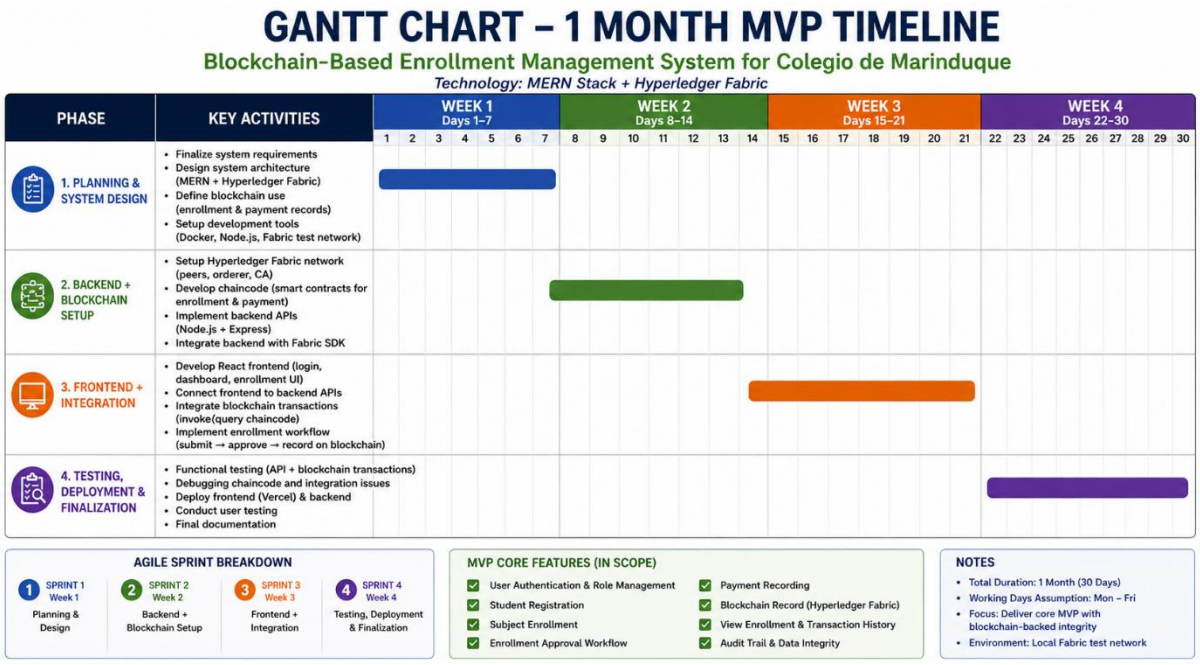

Figure 1: Gantt Chart

The Gantt chart presents a streamlined 1-month Agile timeline for developing the MVP of the Blockchain-Based Enrollment Management System. In Week 1, planning and system design establish the project foundation, including requirements, architecture, and development setup. Week 2 focuses on backend and blockchain development, where the Hyperledger Fabric network, chaincode, and core APIs are implemented and integrated. In Week 3, the React frontend is developed and connected to the backend, enabling key workflows such as enrollment and blockchain transaction processing. Finally, Week 4 covers testing, deployment, and finalization, ensuring the system is functional, integrated, and ready for evaluation. Overall, the timeline emphasizes efficient, iterative development while prioritizing core features and real blockchain integration within a one-month MVP scope.

System Feasibility Analysis

The proposed Blockchain-Based Enrollment Management System is feasible across technical, operational, economic, legal, and schedule aspects, particularly with the adoption of Hyperledger Fabric as a permissioned blockchain framework. From a technical perspective, the system utilizes a MERN stack integrated with Hyperledger Fabric, which supports secure, scalable, and enterprise-grade blockchain applications through components such as peers, ordering services, and chaincode (smart contracts). Although Fabric introduces additional complexity compared to simple hashing approaches, its modular architecture and availability of development tools (e.g., Docker and Fabric SDK) make implementation achievable within an MVP scope. In terms of operational feasibility, the system aligns with existing institutional workflows by supporting key user roles—students, registrar, cashier, and administrators—while automating processes such as registration, enrollment, approval, and payment recording, thereby reducing manual workload and improving efficiency.

From an economic standpoint, the project remains cost-effective by leveraging open-source technologies, including Hyperledger Fabric, and free-tier or low-cost cloud services for deployment. While blockchain infrastructure may require additional setup effort, it does not incur licensing costs, making it sustainable for academic institutions. Regarding legal and ethical feasibility, the system complies with the Data Privacy Act of 2012 (RA 10173) by implementing a permissioned blockchain environment, role-based access control, and a hybrid data storage approach where sensitive personal data are stored off-chain while blockchain records maintain integrity and traceability. Finally, the project is schedule-feasible, as the MVP can be completed within a one-month Agile development cycle by focusing on core features and utilizing pre-configured Hyperledger Fabric test networks. Overall, these factors confirm that the system is practical, secure, and suitable for implementation in a higher education environment.

Conclusion of Proposal

This project proposes a blockchain-integrated enrollment management system that addresses critical inefficiencies in traditional enrollment processes. By combining microservices architecture and blockchain technology, the system ensures:

- Faster enrollment processing
- Secure and tamper-proof records
- Transparent and traceable transactions
- Scalable and modern system architecture

This study fills a significant research and implementation gap in lo`cal higher education institutions by providing a practical, secure, and user-centered enrollment solution.